Data Analysis with PandasAI, the Generative AI Python Library

With PandasAI, you can turn complex queries into simple conversations, unlocking insights that reshape your approach to data analysis. Let's see how...

Business users and data analysts can now become more productive at work than ever before, thanks to OpenAI's ChatGPT artificial intelligence tool. This new technology's capacity to facilitate the creation of SQL queries is a significant perk for business analysts. For years, business analysts have used SQL to query and analyse data. Learning and writing SQL is now easier than the conventional approach, thanks to the usage of AI technology, which can swiftly and automatically generate, edit, debug, and optimise SQL queries.

Python Pandas is an open source toolkit that gives data scientists and analysts the ability to manipulate and analyse data. In the preprocessing stage of machine learning and deep learning, the Pandas library is particularly popular. Now, with generative AI capabilities, the PandasAI module enables conversational data frame operation features for queries to the record set of a DataFrame.

Conversational query

PandasAI provides great strength to Pandas. A great amount of time is usually required to prepare a data set for final analysis. With the help of the PandasAI generative tool, it is possible to initiate the desired data analysis task on the raw data set with minimal preprocessing.

As PandasAI works with Pandas like a supplementary tool, one can apply generative tools on Panda's DataFrames and get back resultant DataFrames. OpenAI's PandasAI can engage in dialogue with a machine to get the desired results in DataFrame format without the need for writing lengthy queries or graphical Python codes.

Use of PandasAI: Installation and setup

The first task is to install PandasAI using the *pip install* command from the command line:

```
C:\mypython> pip install pandasai
```

Once installation is complete, the system is ready for action. Here, the Google Colab platform is used to demonstrate the role of PandasAI in data analysis. Start by calling *https://colab.research.google.com/* and then import the necessary modules as shown here:

```
import pandas as pd
import numpy as np
from pandasai import PandasAI
from pandasai.llm.openai import OpenAI
```

To upload the required input CSV file from the local directory, write down these file uploading instructions:

```
from google.colab import files
uploaded = files.upload()
```